

Delhivery Network Intelligence Dashboard

Graph-Based ETA Prediction and Logistics

Network Optimization

Executive Summary

This project develops a graph-based logistics intelligence framework for Delhivery's transportation network using shipment-level analytics, graph learning, Node2Vec embeddings, and GraphSAGE embeddings. The solution identifies bottleneck hubs, chronic delay corridors, and improves ETA prediction accuracy.

1. Problem Statement

Traditional ETA systems rely heavily on route and distance information and often fail to capture network-level bottlenecks and congestion effects. This project aims to model Delhivery's logistics network as a graph and improve operational decision-making.

2. Dataset Description

The dataset contains shipment movement records including source hubs, destination hubs, route types, travel times, distances, and operational metrics. After cleaning, over 143,000 shipments were analyzed.

3. Methodology

- Part 1: EDA and Data Quality Assessment
- Part 2: Logistics Network Construction
- Part 3: Hub Analytics
- Part 4: Bottleneck Detection
- Part 5: Delay Corridor Analysis
- Part 6: ETA Prediction
- Part 7: Node2Vec Embeddings
- Part 8: GraphSAGE Embeddings
- Part 9: FTL vs Carting Framework

4. Network Analytics

A directed graph was constructed where nodes represent logistics hubs and edges represent shipment corridors. Centrality metrics and network statistics were used to identify critical infrastructure within the network.

5. ETA Prediction

Multiple machine learning models were trained and evaluated including Random Forest, XGBoost, LightGBM, Node2Vec-enhanced models, and GraphSAGE-enhanced models. Performance was

measured using MAE, RMSE, R^2 , and Within $\pm 15\%$ Accuracy.

6. Results

OSRM Baseline MAE: 206.56 seconds
Random Forest MAE: 37.29 seconds
Node2Vec MAE: 36.56 seconds
GraphSAGE Within $\pm 15\%$ Accuracy: 59.6%
Network Hubs Analyzed: 1,601
Chronic Delay Corridors Identified: 20

7. Business Impact

The framework enables bottleneck identification, corridor optimization, ETA improvement, SLA monitoring, and route selection decisions. The project provides both predictive intelligence and operational recommendations.

8. Conclusion

Graph-based machine learning significantly improves logistics network visibility and ETA prediction. Node2Vec and GraphSAGE successfully capture network structure and provide valuable operational insights beyond traditional models.

References

NetworkX Documentation
Scikit-Learn Documentation
PyTorch Geometric Documentation
Node2Vec Research Paper
GraphSAGE Research Paper